

Cartesia Voice Agent Backend Architecture & API Specification

Complete technical specification for Inbound/Outbound AI Cold Calling, Powered Exclusively by Cartesia Voice Agent (cartesia.ai), CRM Database Schemas, and REST/WebSocket APIs.

DOCUMENT VERSION	VOICE ENGINE	TELEPHONY ENGINE	RELEASE DATE
3.0 (Cartesia Dedicated)	Cartesia Voice Agent API	Twilio Media Streams / SIP	August 2026

1 Executive Summary & Cartesia Voice Agent Model

CallX AI is an enterprise AI Voice Cold Calling & CRM Platform powered exclusively by **Cartesia Voice Agent** (<https://cartesia.ai/>) for all conversational intelligence, speech recognition, real-time turn-taking, and hyper-realistic sonic voice generation.

Why Cartesia Voice Agent?

- **All-in-One Voice Agent Architecture:** Eliminates fragmented multi-vendor latency by consolidating VAD (Voice Activity Detection), STT, Conversational Context, Tool Execution, and Sonic TTS into a unified Cartesia Voice Agent session.
- **Ultra-Low Latency (~90ms TTS / <350ms Total Roundtrip):** Industry-leading response speeds creating genuine, human-like voice conversations without awkward pauses.
- **Natural Turn-Taking & Interruption (Barge-in):** Cartesia Voice Agent natively detects when the customer speaks mid-sentence, stops speaking immediately, and listens attentively.
- **Dynamic Tool Calling (Function Execution):** Cartesia Voice Agent directly invokes backend webhooks to update lead scores, log qualification answers, trigger follow-up emails, or initiate live warm transfers.

2 Cartesia Voice Agent Calling Pipeline

```
+-----+
| UNIFIED CARTESIA VOICE AGENT INTEGRATION ARCHITECTURE | +-----+
+-----+ [ Caller / Customer: Mobile or Landline ]
| (Bidirectional Audio Stream – μ-law 8kHz via Twilio or PCM 16kHz via WebRTC) v [ Twilio Voice Media
Stream / Telephony Gateway ] | v (WebSocket Audio Bridge) [ CallX Backend Gateway (Node.js / Python)
] | v (Cartesia Voice Agent WebSocket: wss://api.cartesia.ai/agents/v1/stream) +-----+
+-----+ | CARTESIA VOICE AGENT
ENGINE (cartesia.ai) | | | 1. Streaming Audio Ingestion & Natural Voice Activity Detection (VAD) |
| 2. Turn-Taking & Interruption (Barge-In) Controller | | 3. Conversational Reasoning with Configured
System Persona & Scenario Scripts | | 4. Tool Calling Engine (Invokes CallX CRM webhook for Hot Lead
tagging & email triggers) | | 5. Cartesia Sonic Generative Text-to-Speech Engine (Emma / David /
Sarah / Michael) | +-----+
+-----+ | v (Synthesized Voice Audio Chunks Stream) [ Caller hears seamless, human-like voice
responses in real time ] +-----+
+-----+
```

2.1. Inbound vs. Outbound Cold Calling Workflows

LIFECYCLE STAGE	OUTBOUND COLD CALLING CAMPAIGN	INBOUND RECEPTION & SUPPORT AGENT
Trigger Mechanism	Manual click, scheduled follow-up date, or Batch Campaign (1-10 concurrent lines).	Customer dials dedicated company Twilio/SIP phone number.
Greeting & Opening	Agent speaks first with scenario-defined <code>openingGreeting</code> and value pitch.	Agent answers immediately with company reception greeting & business hours verification.
Qualification & Triage	Asks defined <code>qualifyingQuestions</code> to score lead (Hot, Warm, Cold, Lost).	Identifies caller's intent, answers inquiries from FAQ knowledge base, and logs details.
Handoff & Next Steps	Cartesia Agent invokes backend webhook to dispatch dynamic HTML follow-up email & tag CRM.	Resolves query, books a callback appointment, or executes a live warm transfer to a human agent.

3 Database Schemas & Data Models (PostgreSQL)

3.1. Master Customers & Leads Table (`customers`)

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
<code>id</code>	BIGSERIAL	PRIMARY KEY	Unique customer ID
<code>full_name</code>	VARCHAR(150)	NOT NULL	Contact Full Name
<code>company_name</code>	VARCHAR(150)	DEFAULT ''	Company / Organization Name
<code>job_title</code>	VARCHAR(100)	DEFAULT ''	Title (CEO, Project Manager, GC)
<code>email</code>	VARCHAR(255)	INDEX	Email address
<code>phone</code>	VARCHAR(30)	NOT NULL, INDEX	Primary phone number in E.164 format
<code>website</code>	VARCHAR(255)	DEFAULT ''	Company website URL
<code>address</code> , <code>city</code> , <code>state</code> , <code>country</code>	VARCHAR	NULLABLE	Location metadata
<code>company_type</code>	VARCHAR(50)	DEFAULT 'General'	GC, Developer, Trade, Startup, Agency
<code>lead_status</code>	VARCHAR(50)	DEFAULT 'New'	New, Contacted, Qualified, Won, Lost
<code>lead_priority</code>	VARCHAR(50)	DEFAULT 'Warm'	Hot, Warm, Cold
<code>lead_quality</code>	VARCHAR(50)	DEFAULT 'Good'	Excellent, Good, Average, Poor
<code>last_contact_result</code>	VARCHAR(100)	DEFAULT 'Interested'	Interested, Call back, Meeting booked, No answer
<code>next_follow_up_date</code>	TIMESTAMPTZ	NULLABLE	Scheduled date for next follow-up call
<code>status</code>	VARCHAR(20)	DEFAULT 'Active'	Active / Deactive
<code>reason_for_contact</code>	TEXT	DEFAULT ''	Initial context & sales outreach reason
<code>tags</code>	VARCHAR[]	DEFAULT '{}'	Custom tags list
<code>created_at</code> , <code>updated_at</code>	TIMESTAMPTZ	DEFAULT NOW()	Timestamps

3.2. Customer Notes & Documents Tables

TABLE	COLUMNS	DESCRIPTION
<code>customer_notes</code>	<code>id</code> (PK), <code>customer_id</code> (FK), <code>content</code> (TEXT), <code>author</code> (VARCHAR), <code>created_at</code>	Internal and Cartesia Voice Agent automatic conversation notes.
<code>customer_documents</code>	<code>id</code> (PK), <code>customer_id</code> (FK), <code>name</code> , <code>size</code> , <code>type</code> (Quote/Proposal/Invoice), <code>file_url</code> , <code>upload_date</code>	Attachments stored in AWS S3 / Cloudflare R2.

3.3. Call Logs & Cartesia Transcripts Table (`call_logs`)

COLUMN	TYPE	DESCRIPTION
<code>id</code>	UUID (PK)	Unique call log identifier
<code>customer_id</code>	BIGINT (FK, Nullable)	Associated customer reference
<code>direction</code>	VARCHAR(20)	'inbound' or 'outbound'
<code>phone</code>	VARCHAR(30)	Destination or caller phone number
<code>status</code>	VARCHAR(50)	Completed, Failed, Queued, Upcoming, Callback Requested
<code>duration_seconds</code>	INTEGER	Total call duration in seconds
<code>cartesia_agent_id</code>	VARCHAR(100)	Cartesia Voice Agent ID used
<code>assignee</code>	VARCHAR(100)	Voice Persona (e.g. 'Emma (Cartesia Sonic)')
<code>call_date</code> , <code>call_time</code>	DATE / TIME	Date and start time
<code>recording_url</code>	TEXT (Nullable)	Call audio recording file URL
<code>transcript</code>	JSONB (Nullable)	Turn-by-turn conversational transcript from Cartesia
<code>outcome_status</code>	VARCHAR(50)	Closed/Won, Qualified, Callback Requested, No Answer, Lost

3.4. AI Scenarios & Cartesia Agent Personas (ai_scenarios)

COLUMN	TYPE	DESCRIPTION
id	VARCHAR(50) (PK)	Unique scenario ID
name	VARCHAR(150)	Scenario title (e.g. Inbound Sales Reception, Cold B2B Outreach)
category	VARCHAR(100)	Sales & Outreach, Support, Follow-Up
opening_greeting	TEXT	Spoken hook when call connects
pitch_summary	TEXT	Contextual prompt guidelines for Cartesia Voice Agent
qualifying_questions	JSONB (Array)	Questions Cartesia Agent must evaluate
action_on_interest	VARCHAR(255)	Workflow action when customer expresses high interest
cartesia_voice_id	VARCHAR(100)	Cartesia Voice ID (e.g. Sarah, David, Emma, Michael)
voice_speed	FLOAT	Speed multiplier (0.8x to 1.5x)
voice_tone	VARCHAR(50)	Professional, Friendly, Direct, Casual

4 RESTful & WebSocket API Specification

4.1. Cartesia Voice Agent Dispatch & Campaigns

POST /api/v1/calls/dispatch-single

Initiate an immediate or scheduled outbound cold call via Cartesia Voice Agent.

Request Body:

```
{
  "customerId": 108,
  "scenarioId": "sc_b2b_outreach",
  "cartesiaVoiceId": "sarah",
  "scheduledAt": null
}
```

Response (202 Accepted):

```
{ "status": "Queued", "callSessionId": "cs_cartesia_8819", "targetPhone": "+16045550199" }
```

POST /api/v1/calls/launch-batch-campaign

Launch high-volume cold calling campaign across 1 to 10 concurrent lines.

Request Body:

```
{
  "customerIds": [101, 102, 103, 104, 105],
  "scenarioId": "sc_b2b_outreach",
  "concurrentLines": 5
}
```

WS /ws/cartesia-bridge/:callSessionId

Full-duplex WebSocket Audio Stream between Telephony Media Streams & Cartesia Voice Agent API.

POST /api/v1/cartesia/tool-callback

Webhook endpoint invoked by Cartesia Voice Agent when executing CRM tools (e.g. Update Lead Status, Schedule Callback, Send Follow-up Email).

Request from Cartesia:

```
{
  "tool": "update_lead_qualification",
  "customerId": 108,
  "qualification": { "status": "Qualified", "priority": "Hot", "notes": "Interested in 5 enterprise lines" }
}
```

4.2. Leads & CRM Endpoints

GET `/api/v1/customers`

Get paginated leads with full search and filtering (by country, state, city, leadStatus, leadPriority, custom tags).

POST `/api/v1/customers`

Create lead record with qualification metrics.

GET `/api/v1/customers/:id`

Fetch complete customer dossier including multi-author notes and document attachments.

4.3. Automated Email Follow-ups

POST `/api/v1/emails/send`

Dispatch single or bulk post-call emails using HTML templates with customer variable injection (`{name}` , `{company}` , `{agent}` , `{date}`).

5 Production Architecture & Cartesia Agent SDK Implementation

Production Architecture Stack:

- **Backend Core:** Node.js (NestJS / TypeScript) or Python (FastAPI / AsyncIO).
- **Voice Agent SDK:** [@cartesia/cartesia-js](#) (Cartesia Voice Agent WebSocket API).
- **Telephony Gateway:** Twilio Voice Media Streams (Bidirectional 8kHz μ -law WebSocket) or LiveKit.
- **Database:** PostgreSQL 16+ with Prisma ORM / SQLAlchemy.
- **Concurrency & Queue Management:** Redis + BullMQ (Handles 1-10 concurrent lines dialing and retry queues).
- **Storage:** AWS S3 / Cloudflare R2 (for call recordings and PDF documents).

5.1. Cartesia Voice Agent WebSocket Bridge (Node.js Reference Implementation)

```
import WebSocket from 'ws'; import Cartesia from '@cartesia/cartesia-js'; const cartesia = new
Cartesia({ apiKey: process.env.CARTESIA_API_KEY }); // Handle Twilio Telephony Media Stream
Connection export function setupCartesiaVoiceBridge(twilioWs, callSession) { // Initialize Cartesia
Voice Agent Session const cartesiaAgentWs = cartesia.agents.connect({ agentId:
callSession.cartesiaAgentId, sampleRate: 8000, encoding: "pcm_mulaw" }); let streamSid = null; // 1.
Forward Audio from Caller (Twilio) to Cartesia Voice Agent twilioWs.on('message', (message) => {
const data = JSON.parse(message); if (data.event === 'start') { streamSid = data.start.streamSid; }
else if (data.event === 'media') { const audioBuffer = Buffer.from(data.media.payload, 'base64');
cartesiaAgentWs.sendAudio(audioBuffer); } }); // 2. Stream Synthesized Voice Audio from Cartesia back
to Caller cartesiaAgentWs.on('audio', (audioChunk) => { if (streamSid) {
twilioWs.send(JSON.stringify({ event: 'media', streamSid: streamSid, media: { payload:
Buffer.from(audioChunk).toString('base64') } })); } }); // 3. Handle Cartesia Voice Agent Tool Calls
(CRM Auto-Update) cartesiaAgentWs.on('tool_call', async (toolCall) => { if (toolCall.name ===
'update_customer_lead') { await updateCustomerCRM(callSession.customerId, toolCall.arguments); } });
}
```